# Lightweight Spectral–Spatial Feature Extraction Network Based on Domain Generalization for Cross-Scene Hyperspectral Image Classification

Ying Cui, Longyu Zhu, Chunhui Zhao, Liguo Wang, *Member, IEEE*, and Shan Gao

***Abstract*— The classification of land cover material based on hyperspectral image (HSI) has important research significance. Owing to the high cost of obtaining labeled samples and insufficient training samples, the research of cross-scene HSI classification (CS-HSIC) is receiving more and more attention. At present, the performance of the feature extraction module of CS-HSIC is relatively poor, and the number of training parameters is usually large. To fill the shortcomings of domain generalization (DG) methods and reduce the number of parameters, we propose a lightweight DG network with an attention-assisted cascaded bottleneck (ACB), and it adopts a lightweight bottleneck and multiattention design. This model is adept at extracting domain invariant information contained in the source domain (SD), and it may be flexibly embedded into other models. The experimental results show that our network has good classification accuracy and DG ability when the number of training samples is a little small. As a feature extraction subnetwork, it can improve the performance of the original model or reduce the required resources. The code will be available at https://github.com/zhulongyu1234/ACB.**

***Index Terms*— Attention mechanism, cascaded bottleneck, cross scene, domain generalization (DG), hyperspectral image (HSI) classification.**

## I. Introduction

With the rapid development of hyperspectral image (HSI) science and technology, data analysis of HSI has become a new research direction. It has many application fields such as geological exploration, environmental monitoring, and military reconnaissance [1], [2], [3], and ground object classification is one of its core applications. HSI typically comprises hundreds of continuous narrow bands, which contain rich spatial–spectral information [4]. However, adjacent bands correlation may lead to the Hughes phenomenon in the classification process [5]. In addition, in practical applications, the shortage of labeled training samples may lead to overfitting problems [6].

For the problem of small samples, some researchers have proposed semi-supervised learning solutions by generating pseudo labels [7], [8], but these are highly dependent on the quality of generated pseudo labels. Some researchers also proposed to study more advanced few-shot learning methods to alleviate the dependence on training samples [9], [10], [11]. In recent years, cross-scene HSI classification (CS-HSIC) has become a hot research topic. The significance of CS-HSIC is to use source scenes with sufficient labeled samples and promote the classification of target scenes with insufficient or even no labeled samples, so as to relieve the dependence on training samples of target scenes [12].

Traditional machine learning is based on the assumption that the training and testing sets obey the same distribution. However, the shift of feature space caused by atmosphere and illumination, and the difference of feature space caused by different sensors often break this assumption [13], [14], [15]. In cross-scene tasks, the distribution of different scenes will vary due to changes in the environment and other conditions. Therefore, the common single-scene algorithm applied directly to the cross-scene task often cannot get satisfactory performance. In order to solve the problem of distribution mismatch in cross-scene tasks, researchers have proposed many domain adaptive (DA) methods [16], [17], [18]. By adding the data from the target domain (TD) and combining it with the appropriate DA strategy during the training process, the distribution of the model will be as close as possible to TD. However, most DA methods are based on the assumption that the classes of SD and TD are the same. When there are new classes in TD, classification performance is significantly reduced. Aiming at the situation where SD and TD have inconsistent classes, Zhang et al. proposed the framework of graph information aggregation cross-domain few-shot learning (Gia-CFSL) by combining graph neural network (GNN) and graph alignment [19]. In addition, the research of DA methods extends to the unsupervised branch [20], [21], because there may not be labels of data from TD in practical applications.

Cross-scene methods based on DA are more challenging than that of single-scene methods. However, cross-scene methods based on DG are more difficult than those based on DA,

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and they also have more practical significance and application prospects. Recently, Zhang et al. [22] first proposed the DG model in the CS-HSIC task and proposed a single-source domain expansion network (SDEnet) based on generative adversarial learning. In addition, they combined language knowledge to assist visual representation alignment and proposed a language-aware generalization network (LDGnet) [23]. At present, there are few DG studies in the field of CS-HSIC. Due to its great significance to practical application, more researchers put forward their own schemes for the DG method of CS-HSIC.

Since there is severe information redundancy between bands, and the spatial correlation between each pixel is not very strong, so there is not much spectral–spatial information that can be used for classification. In cross-scene tasks, the information of HSI-specific scenes needs to be discarded, and the public information between scenes is even less, so its feature extraction process may not require too many parameters to store domain invariant information between scenes. Convolutional neural networks (CNNs) are often used in CS-HSIC because of their strong feature extraction capability. Most of the existing methods are implemented by superimposing multiple large-size convolution layers and combining favorable cross-scene strategies, while few methods involve lightweight design. However, large-scale CNNs will increase the spatial complexity and require more computing resources, and complex cross-scene strategies will increase the time complexity and consume more time costs. At present, the existing methods tend to focus more on improving the overall classification performance of the model than on reducing complexity and saving training costs, so research on lightweight models is an important direction. In addition, although CNNs have outstanding advantages in feature extraction, some designs are not good at extracting domain-invariant information for specific cross-scene tasks. More importantly, there is a lot of redundant information in hyperspectral data, and CNNs cannot well distinguish the importance and relevance of each spectral band and spatial location. By distinguishing the importance and correlation, the interference of useless information on classification can be greatly avoided.

In view of the aforementioned issues, this article proposes a lightweight domain generalization (DG) network with an attention-assisted cascaded bottleneck (ACB) to learn common information between scenes. It is a lightweight feature extraction network that does not incorporate complex cross-scene strategies, reducing the number of parameters and computation amount under good classification accuracy. Besides, if the classification performance needs to be further improved in practical applications, it can be flexibly embedded into other appropriate cross-scene classification models as an independent module. The primary contributions of this article are summarized as follows.

1) In order to ensure that the feature extraction process has good domain invariant information extraction ability and low spatial complexity, we establish a cascaded bottleneck module to extract the shared spectral–spatial features between SD and TD. The bottleneck is a lighter structure than ordinary convolution.

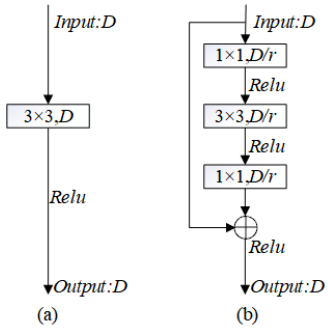



Fig. 1. (a) Structure of the convolution layer. (b) Structure of the bottleneck block.

2) To tackle the challenge that CNNs are not good at distinguishing the correlation and importance of spectral–spatial features, we designed a lightweight multihead self-attention auxiliary module. It can optimize the output features of the cascaded bottleneck module, and further improve the feature learning ability and generalization performance of the model.
3) The proposed feature extraction and optimization network has a simple structure and no complex cross-scene strategy. It can complete cross-scene tasks based on DG under the condition of small time complexity and space complexity. In addition, as a feature extraction subnetwork, it can be flexibly combined with other cross-scene methods to meet various needs in practical applications.

## II. Related Work

### A. Bottleneck

Bottleneck design was first proposed by He et al. [24]. Its core idea is to replace a large convolution kernel with multiple small convolution kernels, so it can reduce the number of parameters in deep neural networks. In this article, a three-layer bottleneck block is used to replace a common convolution layer. As shown in Fig. 1, assume that both the input data and the output data have dimension $D$. The kernel size of the convolutional layer is $3 \times 3$, while the three layers of the bottleneck block are $1 \times 1$, $3 \times 3$, and $1 \times 1$ convolution in turn. The two $1 \times 1$ convolutions are used to decrease and increase dimension, respectively. The dimension change ratio is denoted as $r$. The number of parameters in a convolution layer is shown in (1), while that in a bottleneck block is shown in (2), and the ratio of them is shown in (3). Obviously, when $r > 1$, the number of parameters in the convolutional layer is greater than the bottleneck block. And the larger the $r$, the more obvious the lightening effect of the bottleneck. In addition, the increased depth of the network may have positive implications for accuracy

$$D \times 3 \times 3 \times D = 9D^2 \tag{1}$$

$$D \times 1 \times 1 \times D/r + D/r \times 3 \times 3 \times D/r + D/r \times 1 \times 1 \times D = 9D^2/r^2 + 2D^2/r \tag{2}$$

$$\frac{9D^2}{9D^2/r^2 + 2D^2/r} = \frac{9r^2}{9 + 2r}. \tag{3}$$

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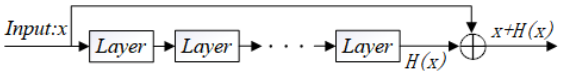



Fig. 2. Diagram of residual structure.

The bottleneck block uses the residual structure, which sums the input data and its output after several layers so that the partial content of the bottleneck layer is linearly contributed by its previous layers, as shown in Fig. 2. For ordinary networks, as the depth of the network deepens, training errors will first decrease and then increase. This phenomenon has nothing to do with overfitting, and it's because the network is difficult to train with being deeper. The proposed residual network can solve the above problems. Even if the network is deeper, the training performance is still good. The residual structure solves the problem of disappearing gradients and exploding gradients, allowing us to train deeper networks while maintaining good information.

### B. Attention Mechanism

Attention mechanism is a computational method that mimics the mechanism of human attention selection [25]. When people read literature, they always pay more attention to the titles, abstracts, figures, and tables, and these places often contain more important information. In deep learning, this kind of focused thinking becomes a module that assigns individual weight values to all elements of a layer's output features. Such attention mechanisms allow neural networks to automatically focus on relevant important information when processing data, thereby improving the performance and generalization ability of the model. People's concern for attention mechanisms remains undiminished. New variants are constantly being proposed for a wide variety of tasks, such as MiniViT [26], MSAFormer [27], and so on.

### C. DA and DG

At present, most CS-HSIC researches adopt the DA method. Traditional DA methods are mainly divided into three types: methods based on subspace learning, methods based on feature selection, and methods based on distribution alignment [28], [29], [30], [31]. Recently, some researchers began to propose the DG model to solve the CS-HSIC task. As shown in Table I, the difference between DG and DA is whether the data from TD can be accessed during training. Owing to the difficulty in obtaining TD data, DG is more challenging than DA, and it is also more promising.

### D. Model Compression

Designing lightweight CNNs while maintaining strong feature extraction capabilities is undoubtedly appealing in various fields. For HSI classification, the research of lightweight deep learning models mainly focuses on single-scene classification. There are currently few lightweight models dedicated to cross-scene tasks. Existing model compression strategies include pruning [32], structural sparsity [33], quantization [34], and distillation [35].

The objective of this article is to construct a lightweight CS-HSIC model with strong feature extraction capability.

TABLE I
COMPARISON BETWEEN DA AND DG

| Learning Paradigm | Training Data | Testing Data |
|---|---|---|
| Domain Adaption | SD,TD | TD |
| Domain Generalization | SD | TD |

Therefore, we use the structural pruning method to compress the model. Remove less important parameters based on their impact on feature extraction performance, such as reducing the number of channels for $3 \times 3$ convolution.

## III. PROPOSED MODEL

### A. Lightweight DG Network With ACB

The flowchart of the lightweight DG network with ACB proposed in this article is shown in Fig. 3. In the training stage, assume that the input HSI dataset is $\boldsymbol{X} \in \mathbb{R}^{B\times H\times W}$, where $H$ and $W$ are the width and height of the spatial dimension, and $B$ is the number of bands of HSI. We take the spatial position of each sample as the center and take the $B \times 12 \times 12$ cube $\boldsymbol{C_{input}}$ as the input of a single sample. We complete the data preprocessing of dimensionality reduction and spatial size transformation with the first bottleneck block, then obtain the output cube with the shape of $D \times 6 \times 6$. After that, we stack $n-1$ bottleneck blocks to extract domain invariant information contained in the SD, where $n$ is the total number of bottleneck blocks. In this article, the output of all bottleneck blocks uses the same dimension $D$. The output of the cascaded bottleneck module is $\boldsymbol{C_{output}} \in \mathbb{R}^{D\times 3\times 3}$, and it will be fed into the lightweight attention module. The output size of the lightweight attention module is the same as the input size. Finally, the prediction probability of the training sample is obtained through the classification head. If we need to embed our proposed network into other cross-scene methods, we only need to parallel a projection head at the location of the classification head. The purpose of the projection head is to change the features of the network into the specified dimensions so that our network can better integrate with other methods. In the test stage, the test samples are input into the model in the same way to get the prediction results of all samples. The structure of the lightweight DG network with ACB is shown in Table II. The number of bottlenecks $n$ is 4 in the table. $B$ and $D$ have the same meaning as those in Fig. 3, while BZ, $l$, FC, and $C$ stand for batch size, the number of attention blocks, the full connection layer, and the number of classes for classification tasks, respectively.

*1) Cascaded Bottleneck Module:* Some CS-HSIC methods directly used classical models in the field of image classification to extract features, such as ResNet [24] and VGG [36]. There are also some methods with new feature extraction modules, but these feature extraction modules are stacked with some common convolution and pooling layers [37], [38]. Their parameter number is relatively large, and the ability to extract domain invariant information is not necessarily good. The bottleneck design cannot only greatly reduce the parameters number by reducing the channel number for $3 \times 3$ convolution, but also increase the depth of the model by replacing a single convolution layer with three convolution layers. The

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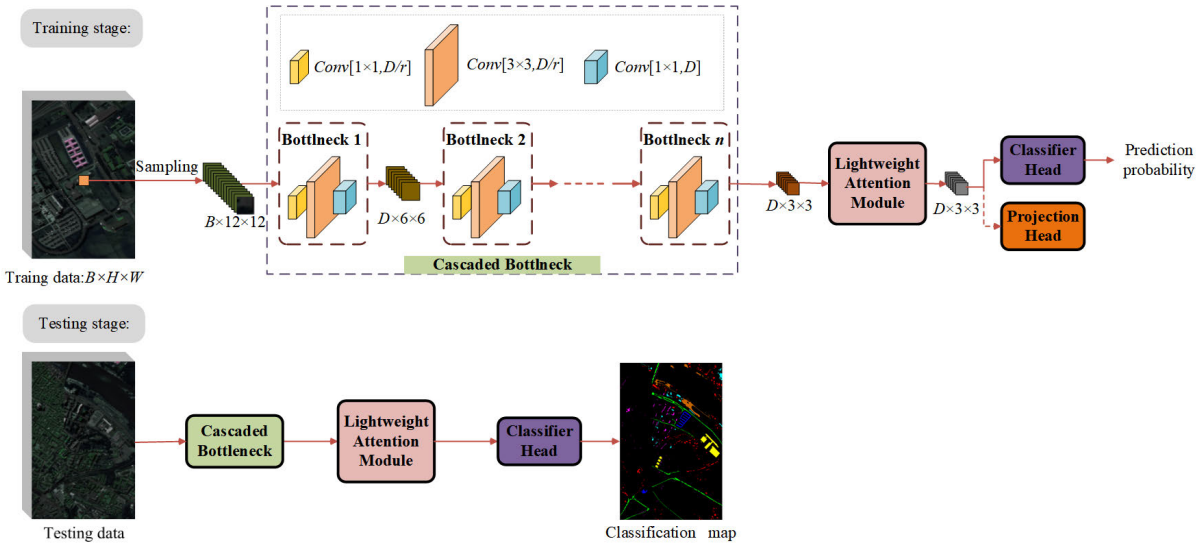



Fig. 3. Flowchart of lightweight DG network with ACB. The network consists of a cascaded bottleneck module and a lightweight attention module.

TABLE II
STRUCTURE OF LIGHTWEIGHT DG NETWORK WITH ACB

| Modules | Operations/blocks | Output dimension |
|---|---|---|
| Input HSI | Sampling | BZ×$B \times 12 \times 12$ |
| Cascaded Bottleneck Module | Bottleneck($D$,2) | BZ×$D \times 6 \times 6$ |
| | Bottleneck($D$,1) | BZ×$D \times 6 \times 6$ |
| | Bottleneck($D$,2) | BZ×$D \times 3 \times 3$ |
| | Bottleneck($D$,1) | BZ×$D \times 3 \times 3$ |
| Lightweight Attention Module | Flatten | BZ×$D \times 9$ |
| | Embedding | BZ×$D \times 10$ |
| | Attention Block×$l$ | BZ×$D \times 10$ |
| | Segment | BZ×$D \times 9$ |
| | Reshape | BZ×$D \times 3 \times 3$ |
| Classifier Head | Flatten | BZ×$9D$ |
| | FC | BZ×$C$ |

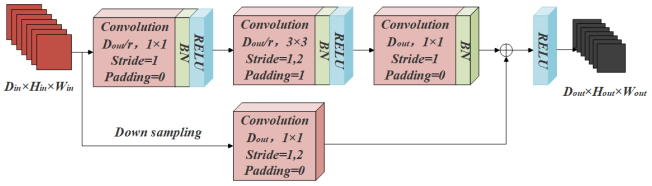



Fig. 4. Flowchart of the bottleneck block.

structure of the bottleneck block is shown in Fig. 4, where $r$ is the reduction multiple of the number of channels for the $3 \times 3$ convolution. In this article, multiple stacked bottleneck blocks are used to extract the domain-invariant information contained in the input data. It ensures the feature extraction ability and reduces the computational resources.

For the cascaded bottleneck module, the input and output dimensions of other bottleneck blocks are the same except for the first bottleneck block, denoted as $D$. Since the input size is relatively large, we select several bottleneck blocks and set the stride of their $3 \times 3$ convolution to 2 to achieve the size reduction. At the same time, the stride of the $3 \times 3$ convolution in the other unselected bottleneck blocks is 1. Generally, the first and third bottleneck blocks are selected. To ensure the size matching of residual connections, we design a parallel subsampled convolution whose stride is equal to that of the $3 \times 3$ convolution for every bottleneck block.

*2) Lightweight Multihead Attention Module:* Since the large spatial resolution of HSI and the serious spectral redundancy problem, the correlation and importance between different spatial locations and bands have an impact on classification. Although the cascaded bottleneck module can extract domain invariant information, the output spectral–spatial features are not properly assigned according to the correlation and importance of bands and spatial locations. To solve the above problems, this article designs a multihead attention lightweight auxiliary module, which can reasonably allocate the weight of each feature to optimize the output spectral–spatial features. The structure is shown in Fig. 5. The attention module first expands the spatial dimension of the input feature and adds the position encoding through the embedding operation to avoid the position information being lost after the attention calculation. The features after the embedding operation are entered into $l$ cascaded multihead attention blocks for feature optimization. Finally, the output features of the attention blocks are segmented and transformed into the input shape for subsequent operations. In this article, $l$ is 1 based on the joint consideration of model performance and computing resources. This model can analyze the input data from multiple perspectives and learn the dependencies of different positions of the feature sequence so that it can fully understand the input features and improve generalization ability. In addition, multiple head calculations can be performed in parallel, which has positive implications for the training speed.

Assuming that $\boldsymbol{X}$ is the input feature of the attention block, when calculating the attention weight, $\boldsymbol{X}$ needs to be multiplied by three weight matrices, respectively, for linear transformation to obtain three different input features, as follows:

$$\begin{aligned} \boldsymbol{W}_q \cdot \boldsymbol{X} &= \boldsymbol{Q} \\ \boldsymbol{W}_k \cdot \boldsymbol{X} &= \boldsymbol{K} \\ \boldsymbol{W}_v \cdot \boldsymbol{X} &= \boldsymbol{V} \end{aligned} \tag{4}$$

where $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, and $\boldsymbol{W}_v$ are the query matrix, key matrix, and value matrix, respectively, which are all implemented by the fully connected layer. Although larger networks

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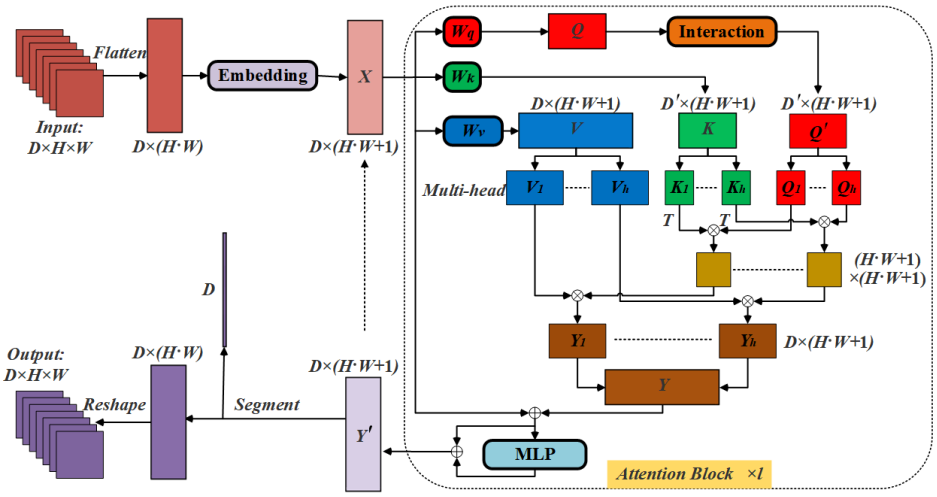



Fig. 5. Flowchart of the lightweight multihead attention module. The input and output features of the module are the same size.

generally perform better, they are also considered to be overparameterized. Inspired by Yang et al. [39], we can prune the structure of the attention module under certain resource constraints, remove the unimportant channels, and retain the most critical channels. For the same dimension, the redundancy of $\boldsymbol{Q}$ and $\boldsymbol{K}$ is much greater than that of $\boldsymbol{V}$ [40]. In this article, we no longer use the same dimensional $\boldsymbol{W_q}$, $\boldsymbol{W_k}$, and $\boldsymbol{W_v}$ as ordinary transformer. By pruning the $\boldsymbol{W_q}$ and $\boldsymbol{W_k}$ dimensions to reassign parameters, the parameters and computation of the attention module are reduced without affecting the performance. In this article, the dimensions of $\boldsymbol{W_q}$ and $\boldsymbol{W_k}$ are half that of $\boldsymbol{W_v}$ to realize the lightweight design of the attention module.

$\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are three different input features. $\boldsymbol{Q}$ needs to first enrich local characterization through an interaction layer to obtain $\boldsymbol{Q}'$. $\boldsymbol{Q}'$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are then divided into $h$ parts, where $h$ is the number of heads of the attention mechanism, and $h$ in this article is 2. Then, calculate the multihead attention output

$$\text{Softmax}\left(\frac{\boldsymbol{Q}_i' \cdot \boldsymbol{K}_i^{\mathrm{T}}}{\sqrt{d}}\right)\boldsymbol{V}_i = \boldsymbol{A}\boldsymbol{V}_i = \boldsymbol{Y}_i \tag{5}$$

where $T$ represents the transpose operation, $i$ represents the $i$th head, $d$ represents the dimensions of $\boldsymbol{Q}_i$ and $\boldsymbol{K}_i$, $\boldsymbol{A}$ represents the attention matrix, and $\boldsymbol{Y}_i$ represents the attention output of the $i$th head. The output of the $h$ head is merged

$$\text{concat}[\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_h] = \boldsymbol{Y} \tag{6}$$

where concat is the concatenation operation of matrices in channel dimension. By connecting two residuals and an MLP layer, we obtain the output $\boldsymbol{Y}'$ of the entire attention block

$$\text{MLP}(\boldsymbol{X} + \boldsymbol{Y}) + \boldsymbol{X} + \boldsymbol{Y} = \boldsymbol{Y}'. \tag{7}$$

Through nonlinear transformations of multiple hidden layers, the MLP layer is able to extract deeper features from the original input, and it can learn more complex feature representations. In this article, the three-layer structure is used.

### B. Application of ACB to Other Methods

This section mainly discusses the importance of the feature extraction module proposed in this article. We replace the feature extraction module in other networks with the ACB model for experimental analysis. In order to effectively integrate ACB networks with other networks, we add the projection head to the end of the network and parallel it with the classification head. For a plain network, the feature dimension is first expanded at the projection head, and one or more fully connected layers are added. The complexity of fully connected layers is quite large under the high dimensions. To reduce the complexity, our network only expands the dimensions without adding additional fully connected layers. Fig. 6 shows diagram of ACB combined with the classical DA methods DeepCoral [41], MRAN [42], DSAN [43], and TSTnet [44]. ACB only replaces the feature extraction module without changing the other parts of these networks, and experimental results and performance analysis will be presented in Section IV-F.

## IV. Experiments

### A. Datasets

The experiments use three common cross-scene datasets, i.e., the Houston dataset, the Pavia dataset, and the Hyrank dataset.

*1) Houston Dataset:* This dataset consists of two scenes in Houston in 2013 [45] and Houston in 2018 [46], which were acquired by different sensors at the University of Houston and its surroundings in different years. The Houston 2013 dataset contains 349 × 1905 pixels and 144 spectral bands, while the Houston 2018 dataset contains 210 × 954 pixels and 48 spectral bands. We extract 48 spectral bands from the Houston 2013 scene to correspond to the number of bands in the Houston 2018 scene and select 210 × 954 pixels of spatial overlap region and seven consistent classes for cross-scene research. The Houston 2013 scene is SD and the Houston 2018 scene is TD. The names of their land cover classes and the number of samples are listed in Table III. The pseudocolor images and ground-truth maps are shown in Fig. 7.

*2) Pavia Dataset:* The dataset consists of two scenes, Pavia University and Pavia Center, captured by different sensors. The Pavia University dataset contains 610 × 340 pixels and 103 bands, while the Pavia Center dataset contains 1096 × 715 pixels and 102 bands. In order to keep the number of bands and classes consistent in the Pavia dataset, we discard

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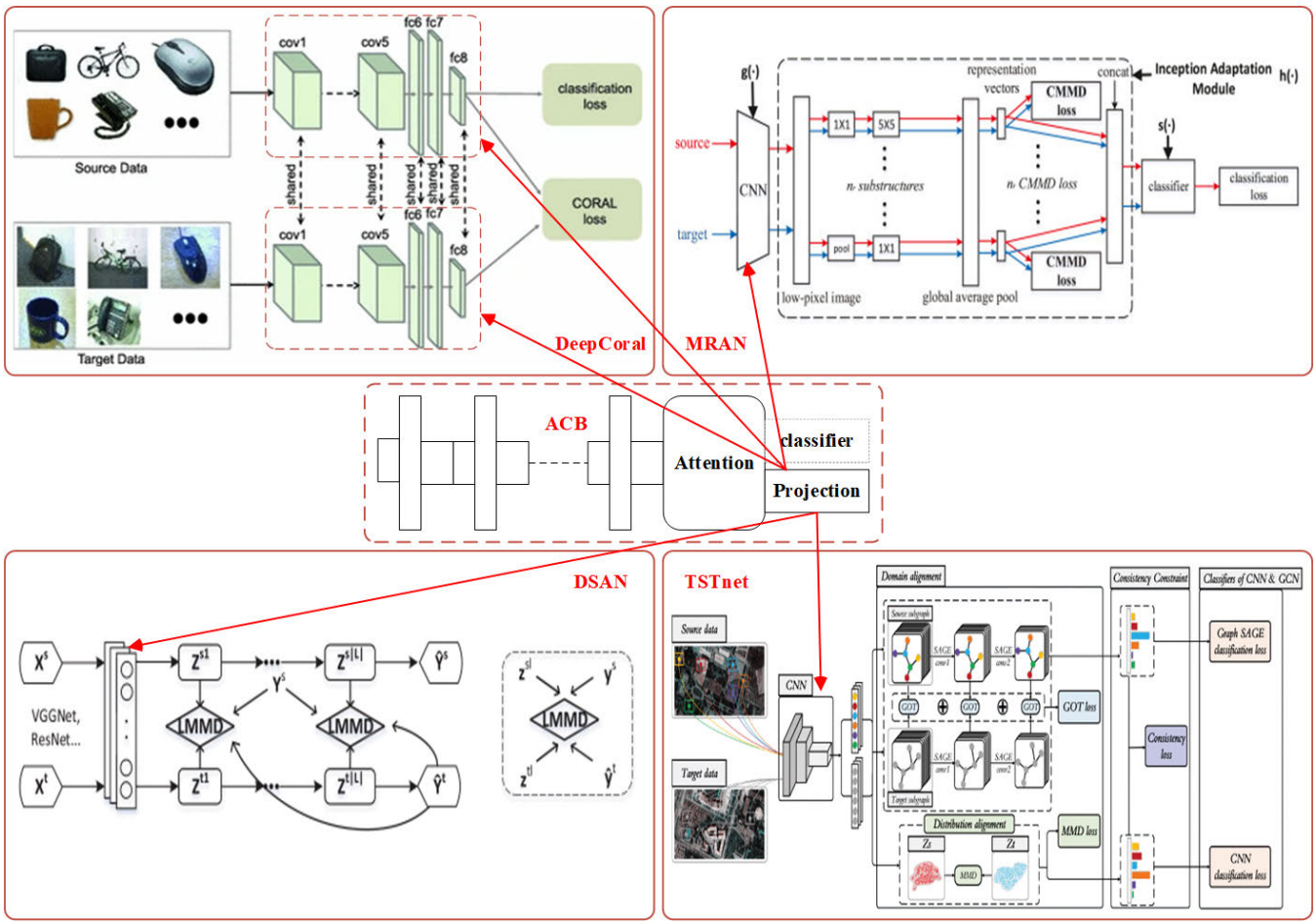



Fig. 6. Diagram of ACB combined with other methods. Inside the red boxes in the top left, top right, bottom left, and bottom right are the flow charts of DeepCoral, MRAN, DSAN, and TSTnet, in turn. They are all quoted from their respective references. Notice that the red line points to where the ACB is embedded.

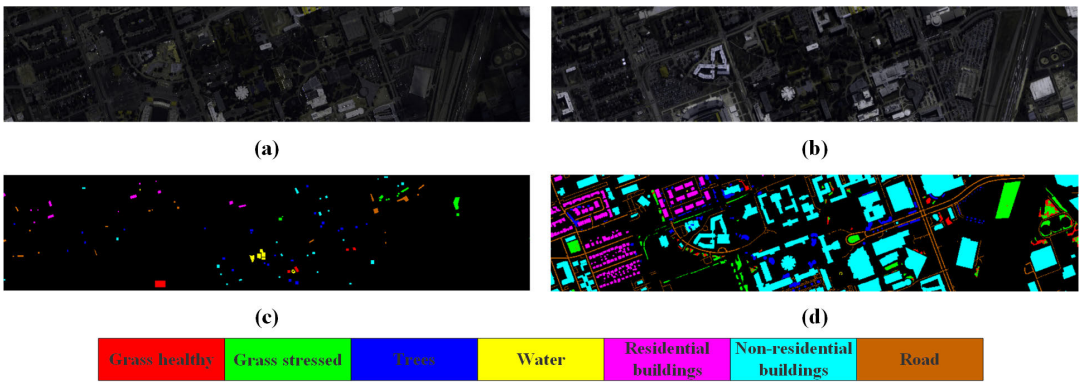



Fig. 7. Pseudocolor images and ground-truth maps of the Houston dataset. (a) Pseudocolor image of Houston 2013. (b) Pseudocolor image of Houston 2018. (c) Ground-truth map of Houston 2013. (d) Ground-truth map of Houston 2018.

the last band of the Pavia University dataset and select seven common classes for the experiment. In the experiments, Pavia University is SD, while Pavia Center is TD. The names of their land cover classes and the number of samples are listed in Table IV. The pseudocolor images and ground-truth maps are shown in Fig. 8.

*3) Hyrank Dataset:* This dataset [47] consists of two scenes, Dioni and Loukia, captured by the same sensor. The Dioni dataset contains 250 × 1376 pixels, while the Loukia dataset contains 246 × 945 pixels, both of which have 176 bands. We choose twelve of their consistent classes for study. In the experiments, Dioni is SD and Loukia is TD. The names and sample numbers of their land cover classes are listed in Table V. The pseudocolor images and ground-truth maps are shown in Fig. 9.

### *B. Experimental Setting*

The experimental environment is Pytorch deep learning framework with 64G running memory, Inter[1] Core[2] i7-10700

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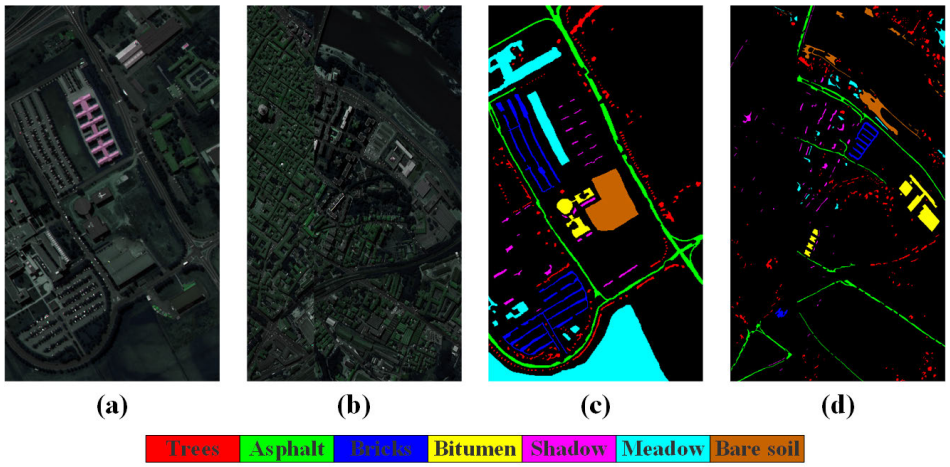



Fig. 8. Pseudocolor images and ground-truth maps of the Pavia dataset. (a) Pseudocolor image of Pavia University. (b) Pseudocolor image of Pavia Center. (c) Ground-truth map of Pavia University. (d) Ground-truth map of Pavia Center.

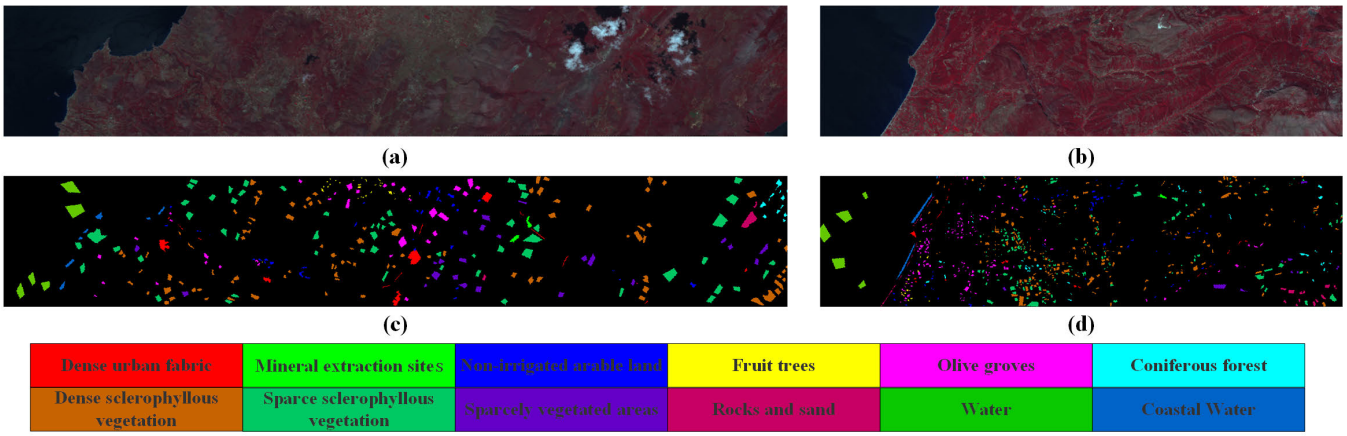



Fig. 9. Pseudocolor images and ground-truth maps of the Hyrank dataset. (a) Pseudocolor image of Dioni. (b) Pseudocolor image of Loukia. (c) Ground-truth map of Dioni. (d) Ground-truth map of Loukia.

TABLE III
CLASSES AND NUMBERS OF SD AND TD FOR THE HOUSTON DATASET

| Class | | Number of samples | |
|---|---|---|---|
| No. | Name | Houston 2013(SD) | Houston 2018(TD) |
| 1 | Grass healthy | 345 | 1353 |
| 2 | Grass stressed | 365 | 4888 |
| 3 | Trees | 365 | 2766 |
| 4 | Water | 285 | 22 |
| 5 | Residential buildings | 319 | 5347 |
| 6 | Non-residential buildings | 408 | 32459 |
| 7 | Road | 443 | 6365 |
| **Total** | | 2530 | 53200 |

TABLE IV
CLASSES AND NUMBERS OF SD AND TD FOR THE PAVIA DATASET

| Class | | Number of samples | |
|---|---|---|---|
| No. | Name | Pavia University(SD) | Pavia Center(TD) |
| 1 | Trees | 3064 | 7598 |
| 2 | Asphalt | 6631 | 9248 |
| 3 | Bricks | 3682 | 2685 |
| 4 | Bitumen | 1330 | 7287 |
| 5 | Shadow | 947 | 2863 |
| 6 | Meadow | 18649 | 3090 |
| 7 | Bare soil | 5029 | 6584 |
| **Total** | | 39332 | 39355 |

CPU@2.9 GHz, and NVIDIA GeForce RTX3060 GPU. Set the patch size of the input cube to 12 × 12 and the batch size to 256. The training samples are randomly selected, and the training sample size is increased by 2 times by random flip enhancement and radiation enhancement. The Adam optimizer with a learning rate of 0.01 is used to iteratively train 300 epochs. The testing set is all samples of TD and is tested once each epoch is trained. Overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa) are used to evaluate the classification performance. The specific definitions and calculation methods of OA, AA,

TABLE V
CLASSES AND NUMBERS OF SD AND TD FOR THE HYRANK DATASET

| Class | | Number of samples | |
|---|---|---|---|
| No. | Name | Dioni(SD) | Loukia(TD) |
| 1 | Dense urban fabric | 1262 | 206 |
| 2 | Mineral extraction sites | 204 | 54 |
| 3 | Non-irrigated arable land | 614 | 426 |
| 4 | Fruit trees | 150 | 79 |
| 5 | Olive groves | 1768 | 1107 |
| 6 | Coniferous forest | 361 | 422 |
| 7 | Dense sclerophyllous vegetation | 5035 | 2996 |
| 8 | Sparce sclerophyllous vegetation | 6374 | 2361 |
| 9 | Sparcely vegetated areas | 1754 | 399 |
| 10 | Rocks and sand | 492 | 453 |
| 11 | Water | 1612 | 1393 |
| 12 | Coastal Water | 398 | 421 |
| **Total** | | 20024 | 10317 |

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TABLE VI
CLASSIFICATION PERFORMANCE OF HOUSTON DATASET WITH DIFFERENT PROPORTIONS OF TRAINING SAMPLES

| Proportion(%) | 10 | 20 | 30 | 40 | 50 | 60 | 80 | 100 |
|---|---|---|---|---|---|---|---|---|
| OA(%) | 72.15 | 75.88 | 77.53 | 79.33 | 79.55 | 79.78 | **80.18** | 79.96 |
| AA(%) | 56.80 | 63.57 | 66.43 | 66.82 | 67.23 | 65.93 | 68.10 | **69.18** |
| Kappa(%) | 54.40 | 59.07 | 62.68 | 64.30 | 64.24 | 64.96 | **65.82** | 65.40 |

TABLE VII
CLASSIFICATION PERFORMANCE OF PAVIA DATASET WITH DIFFERENT PROPORTIONS OF TRAINING SAMPLES

| Proportion(%) | 1 | 2 | 3 | 4 | 5 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|---|---|
| OA(%) | 81.61 | 84.76 | **85.31** | 83.36 | 85.00 | 84.31 | 82.40 | 83.88 |
| AA(%) | 77.55 | 82.37 | **83.58** | 80.44 | 82.29 | 82.30 | 80.01 | 82.50 |
| Kappa(%) | 77.76 | 81.59 | **82.28** | 79.86 | 81.85 | 81.12 | 78.76 | 80.54 |

TABLE VIII
CLASSIFICATION PERFORMANCE OF HYRANK DATASET WITH DIFFERENT PROPORTIONS OF TRAINING SAMPLES

| Proportion(%) | 1 | 2 | 3 | 4 | 5 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|---|---|
| OA(%) | 62.02 | 62.99 | 62.74 | **64.01** | 63.16 | 63.57 | 63.24 | 63.41 |
| AA(%) | 36.51 | 38.53 | 38.67 | 40.35 | 40.51 | 41.08 | **42.05** | 41.63 |
| Kappa(%) | 51.90 | 53.52 | 53.01 | **55.26** | 54.38 | 54.43 | 54.20 | 54.02 |

and Kappa are given as (8)–(10). It measures the degree of agreement between the model prediction results and the actual classification results, which helps us to understand the performance of the model in various situations. The average time required for a single epoch during training is used to represent the time complexity, and the number of training parameters is used to represent the spatial complexity. All experimental results are the average of ten experiments after removing the maximum and minimum values

$$\mathrm{OA} = \frac{\mathrm{IM}}{\mathrm{IN}} \tag{8}$$

$$\mathrm{AA} = \frac{\sum_{k=1}^{K} \frac{\mathrm{IM}_k}{\mathrm{IN}_k}}{K} \tag{9}$$

$$\mathrm{Kappa} = \frac{\mathrm{IM} \times \mathrm{IN} - \sum_{k=1}^{K} \mathrm{IP}_k \times \mathrm{IN}_k}{\mathrm{IN} \times \mathrm{IN} - \sum_{k=1}^{K} \mathrm{IP}_k \times \mathrm{IN}_k} \tag{10}$$

where IM represents the number of all correctly classified test samples, IN represents the total number of test samples, $\mathrm{IM}_k$ represents the number of correctly classified test samples of class $k$, $\mathrm{IN}_k$ represents the total number of test samples of class $k$, and $\mathrm{IP}_k$ represents the number of predicted test samples of class $k$.

### C. Parameter Analysis and Setting

*1) Proportion of Training Samples:* In the training process, the proposed CS-HSIC task alleviates the dependence on TD samples, but it still needs enough labeled samples on SD. In practice, the SD-labeled samples may also be very limited. Therefore, the training samples required by the model should be reduced as much as possible. Tables VI–VIII show the classification performance of the three datasets with different proportions of training samples.

As shown in Table VI, when the proportion of training samples in the Houston dataset is less than 40%, the classification performance is poor. When the proportion is greater than 40%, the classification performance is improved. Considering the difficulty and cost of obtaining training samples, 40% training samples are used for the Houston dataset in subsequent experiments. As shown in Table VII, when the proportion of training samples in the Pavia dataset is less than 2%, the classification performance is poor. When the proportion is greater than 2%, the classification performance fluctuates, and the best classification performance appears at 3%. We balance the sample size and classification accuracy, using 2% training samples for the Pavia dataset. As shown in Table VIII, when the proportion of training samples in the Hyrank dataset is less than 4%, the classification performance is relatively poor. When the proportion is greater than 4%, the classification performance is not improved. In the subsequent experiment, 4% training samples are used for the Hyrank dataset. After comprehensive consideration, the number of training samples of the three datasets is controlled within 1000.

*2) Dimension $D$ of Bottleneck Block:* The cascaded bottleneck module is actually a set of convolutions combined with residual calculation. When the convolution kernel size is fixed, the dimension change of the convolution layer is the main factor affecting the number of parameters. Therefore, the dimension $D$ of bottleneck blocks is a major factor affecting the model space complexity. In this article, the experiment is conducted under the condition of $D \in \{16, 32, 48, 64, 96\}$. The classification performance of the three datasets is shown in Fig. 10. The experimental results show that the classification performance of three datasets does not improve with the increase of the dimension of bottleneck blocks, but fluctuates within a certain range. Considering the computing resources required by the model and the classification performance of the model, the subsequent experiment set $D = 32$ in this article.

*3) Dimension Reduction Multiple $r$ of Bottleneck Blocks:* Bottleneck implements model lightening by reducing the number of channels in $3 \times 3$ convolution, and the multiple $r$ of channel number reduction can affect the lightening degree of the model. The experiments are carried out under the condition of $r \in \{1, 2, 4, 8, 16\}$. The classification performance of the

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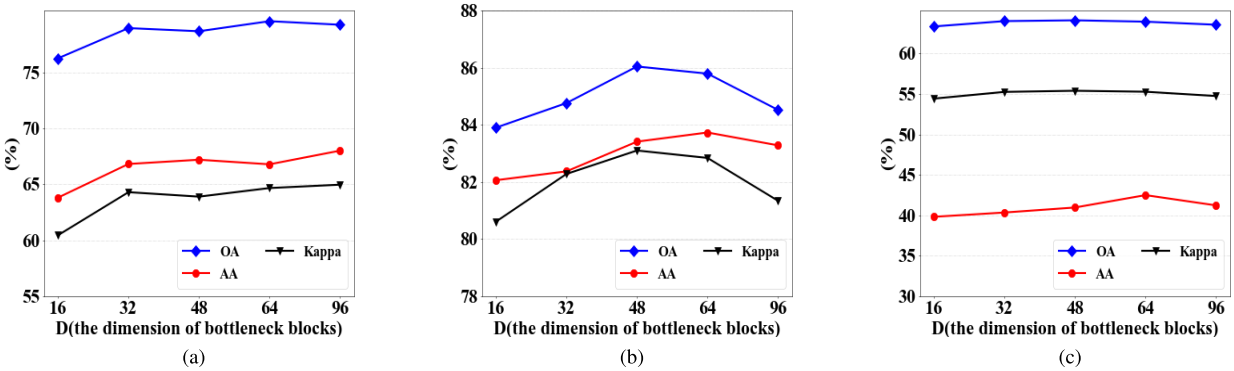



Fig. 10. Classification performance of three datasets under different $D$. (a) Houston. (b) Pavia. (c) Hyrank.

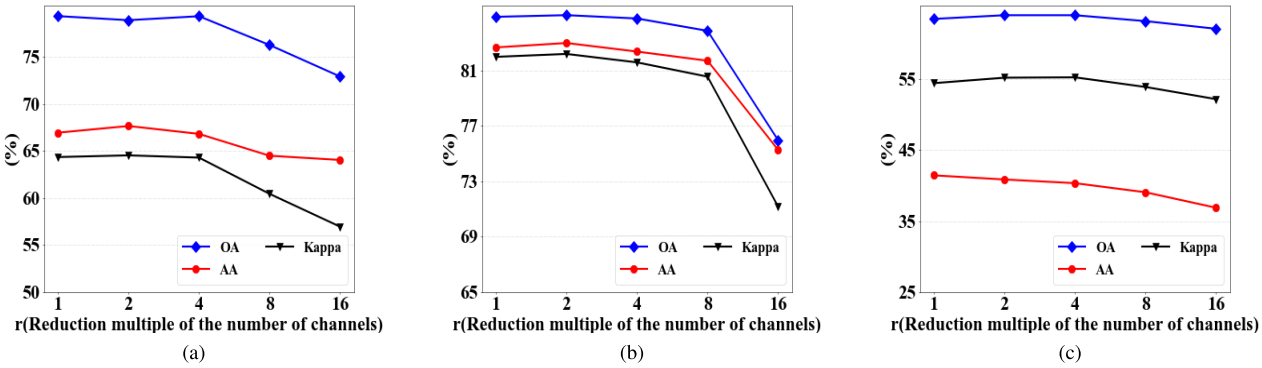



Fig. 11. Classification performance of three datasets under different $r$. (a) Houston. (b) Pavia. (c) Hyrank.

three datasets is shown in Fig. 11. When $r = 1, 2, 4$, the classification performance of the three datasets is quite close. When $r = 8$, the classification performance is reduced to a certain extent. When $r = 16$, the classification performance becomes very terrible. Therefore, we choose $r = 4$ for all three datasets to conduct subsequent experiments.

*4) Number n of Bottleneck Blocks:* In this article, the feature extraction process is completed by cascaded bottleneck blocks, and the model can be lightened by using fewer bottleneck blocks. The experiments are carried out under the condition of $n \in \{2, 3, 4, 5, 6, 7, 8\}$. The classification performance of the three datasets is shown in Fig. 12. The experimental results show that the classification performance of the three datasets does not improve with the increase of bottleneck blocks. The value of $n$ can be selected according to specific tasks. The subsequent experiments are carried out under the condition of $n = 4$.

### D. Ablation Study of Our Network

The lightweight multihead attention module is one of the key components of our network. To evaluate the contribution of this module, we perform an ablation analysis by removing the attention module, and the network after the removal of attention is denoted as ACB (no attention). We conduct ablation experiments on three datasets, respectively, and the results are shown in Table IX.

According to the experimental results, the classification performance with the multihead attention module is increased by 1%–3% on all three datasets, and the amount of parameters is close. This indicates that the designed attention module can optimize the output of the cascaded bottleneck module according to the importance and correlation of spectral–spatial features, and it does not require high computing resources.

### E. Comparative Analysis of Classification Performance

In order to evaluate the classification performance of the proposed lightweight feature extraction network, this article selects the classical feature extraction networks VGGNet11 [36] and ResNet18 [24], the classical DA networks DeepCoral [41], MRAN [42], DSAN [43], and TSTnet [44], the newest DG networks SDEnet [22] and LDGnet [23] as comparison models. Among them, MRAN, DSAN, and DeepCoral are all DA methods in the field of image classification, and MRAN and DSAN use ResNet for feature extraction. Therefore, the feature extraction process of DeepCoral is also replaced by ResNet in this article, instead of the original AlexNet [48]. TSTnet is a DA method for CS-HSIC, with its own feature extraction module. The DA methods here all require data from TD for training, while the other methods only train on SD data. Tables X–XII show the classification performance of different methods on the three datasets. Figs. 13–15 show the classification maps.

Compared with the classical feature extraction networks VGG11 and ResNe18, the ACB network has less confusion in some areas and higher overall classification performance, and best on the Pavia dataset. These three methods show confusion in classes 5 and 6 of the Houston dataset, classes 2–4 of the Pavia dataset, and classes 7–9 of the Hyrank dataset. This

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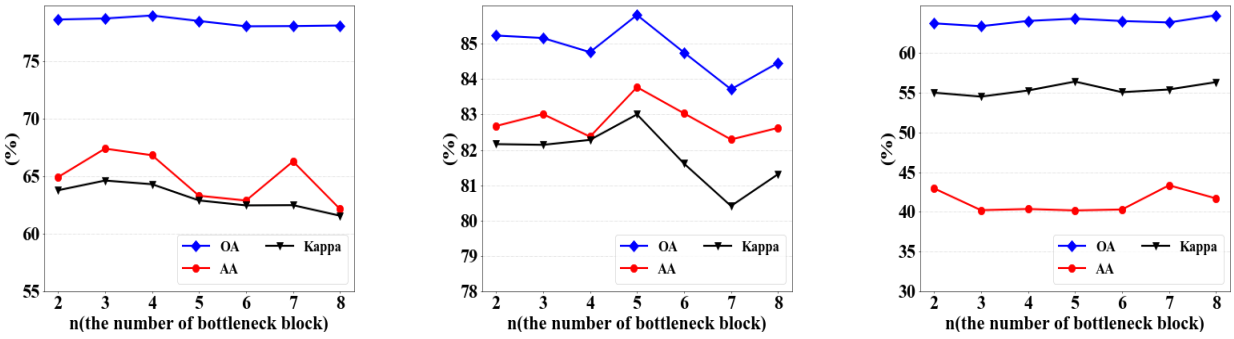



Fig. 12. Classification performance of three datasets under different $n$. (a) Houston. (b) Pavia. (c) Hyrank.

TABLE IX

ABLATION ANALYSIS OF MULTIHEAD ATTENTION MODULE ON THREE DATASETS

| Dataset | **Houston** | | **Pavia** | | **Hyrank** | |
|---|---|---|---|---|---|---|
| Network | ACB(no attention) | ACB | ACB(no attention) | ACB | ACB(no attention) | ACB |
| OA(%) | 78.58 | **79.33** | 81.89 | **84.76** | 62.23 | **64.01** |
| AA(%) | 64.86 | **66.82** | 80.70 | **82.37** | 40.24 | **40.35** |
| Kappa(%) | 63.23 | **64.30** | 78.19 | **81.59** | 52.75 | **55.26** |
| Parameters(M) | 0.01 | 0.02 | 0.01 | 0.02 | 0.02 | 0.02 |

TABLE X

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON HOUSTON DATASET

| Method | VGG11 | ResNet18 | DeepCoral | MRAN | DSAN | TSTnet | SDEnet | LDGnet | ACB |
|---|---|---|---|---|---|---|---|---|---|
| OA(%) | 78.66 | 78.64 | 79.52 | 77.67 | 77.91 | **80.04** | 69.27 | 79.30 | 79.33 |
| AA(%) | 63.38 | 64.50 | 72.84 | 70.12 | 69.88 | **75.50** | 52.13 | 68.53 | 66.82 |
| Kappa(%) | 62.97 | 62.33 | 65.86 | 62.36 | 63.11 | **67.03** | 46.13 | 64.52 | 64.30 |
| Parameters(M) | 42.87 | 10.80 | 10.80 | 11.12 | 10.92 | 7.42 | 1.62 | 34.22 | **0.02** |
| Time(s) | **0.17** | 0.31 | 1.9 | 1.9 | 2.3 | 6.4 | 1.4 | 13 | 0.20 |

TABLE XI

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON PAVIA DATASET

| Method | VGG11 | ResNet18 | DeepCoral | MRAN | DSAN | TSTnet | SDEnet | LDGnet | ACB |
|---|---|---|---|---|---|---|---|---|---|
| OA(%) | 74.80 | 80.28 | 68.32 | 69.03 | 77.23 | 79.85 | 69.96 | 78.00 | **84.76** |
| AA(%) | 68.62 | 77.07 | 66.49 | 65.73 | 72.46 | 75.86 | 69.14 | 75.79 | **82.37** |
| Kappa(%) | 69.47 | 76.18 | 61.77 | 62.53 | 72.45 | 75.61 | 64.3 | 72.99 | **81.59** |
| Parameters(M) | 42.90 | 10.96 | 10.96 | 11.28 | 11.08 | 7.43 | 2.16 | 35.10 | **0.02** |
| Time(s) | **0.16** | 0.26 | 1.9 | 1.9 | 2.1 | 5.2 | 1.3 | 25 | 0.19 |

TABLE XII

CLASSIFICATION PERFORMANCE OF DIFFERENT METHODS ON HYRANK DATASET

| Method | VGG11 | ResNet18 | DeepCoral | MRAN | DSAN | TSTnet | SDEnet | LDGnet | ACB |
|---|---|---|---|---|---|---|---|---|---|
| OA(%) | 60.30 | 64.86 | 66.11 | **67.43** | 66.82 | 66.98 | 54.74 | 59.74 | 64.01 |
| AA(%) | 36.04 | 44.69 | 44.60 | **47.06** | 46.54 | 42.72 | 26.25 | 43.04 | 40.35 |
| Kappa(%) | 49.75 | 56.38 | 58.32 | **60.03** | 59.00 | 59.30 | 42.32 | 50.08 | 55.26 |
| Parameters(M) | 42.96 | 11.18 | 11.18 | 11.50 | 11.30 | 7.47 | 2.89 | 36.28 | **0.02** |
| Time(s) | **0.17** | 0.26 | 1.9 | 2.0 | 2.2 | 5.2 | 1.5 | 33 | 0.19 |

TABLE XIII

CLASSIFICATION PERFORMANCE OF FOUR NEW NETWORKS ON THE HOUSTON DATASET

| Method | DeepCoral(ACB) | MRAN(ACB) | DSAN(ACB) | TSTnet(ACB) |
|---|---|---|---|---|
| OA(%) | **80.46** | 79.69 | 79.00 | 80.37 |
| AA(%) | 72.29 | 69.38 | **72.37** | 70.75 |
| Kappa(%) | 67.15 | 65.00 | 65.05 | **67.35** |
| Parameters(M) | 0.02 | 0.23 | 0.02 | 2.18 |
| Time(s) | 1.8 | 1.9 | 2.1 | 5.8 |

may be because these classes are highly correlated in spectral features, requiring a combination of excellent strategies to make correct judgments. In terms of time complexity, the three feature extraction networks are all small. That's because the

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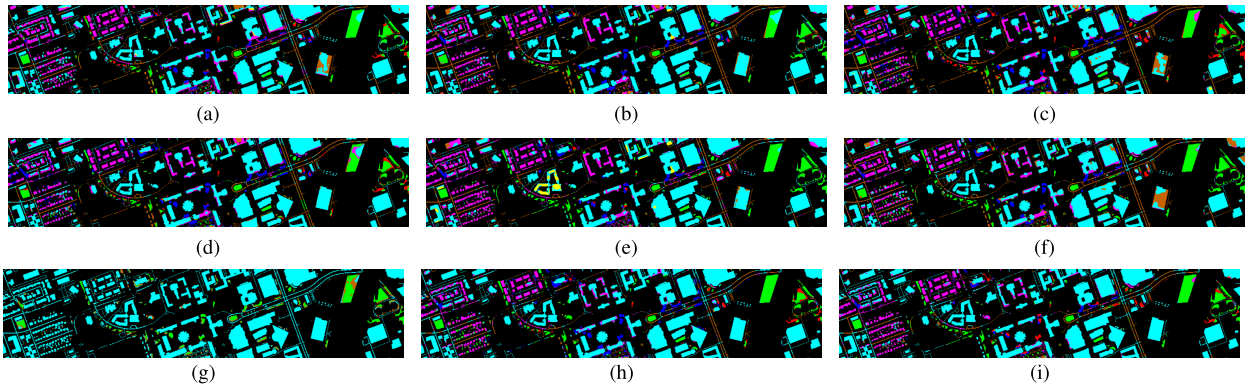


Fig. 13. Classification maps of different methods on the Houston dataset. (a) VGG11. (b) ResNet18. (c) DeepCoral. (d) MRAN. (e) DSAN. (f) TSTnet. (g) SDEnet. (h) LDGnet. (i) ACB.

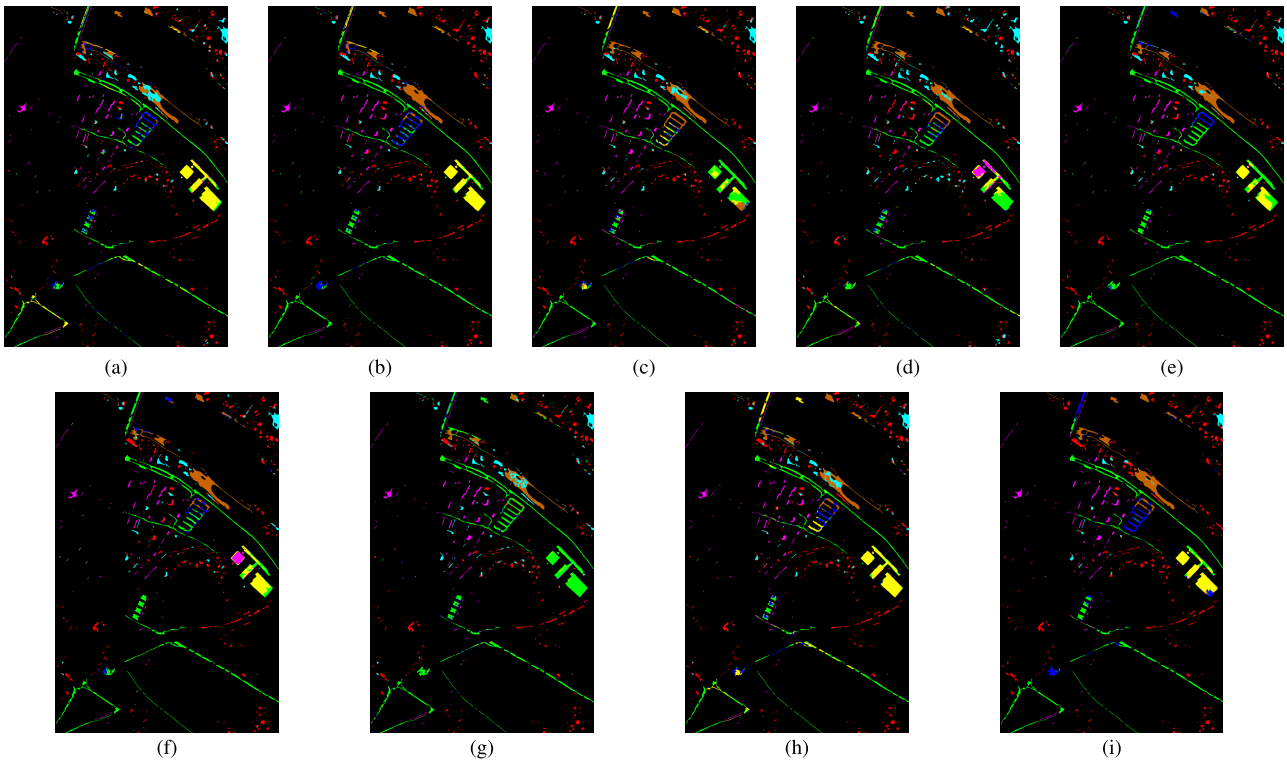


Fig. 14. Classification maps of different methods on the Pavia dataset. (a) VGG11. (b) ResNet18. (c) DeepCoral. (d) MRAN. (e) DSAN. (f) TSTnet. (g) SDEnet. (h) LDGnet. (i) ACB.

TABLE XIV
CLASSIFICATION PERFORMANCE OF FOUR NEW NETWORKS ON THE PAVIA DATASET

| Method | DeepCoral(ACB) | MRAN(ACB) | DSAN(ACB) | TSTnet(ACB) |
|---|---|---|---|---|
| OA(%) | 81.82 | 74.72 | **86.44** | 81.12 |
| AA(%) | 80.61 | 73.23 | **84.75** | 78.25 |
| Kappa(%) | 78.23 | 69.83 | **83.68** | 77.27 |
| Parameters(M) | 0.02 | 0.23 | 0.02 | 2.18 |
| Time(s) | 1.8 | 1.9 | 2.1 | 5.0 |

memory access cost is a key factor affecting the speed of the model [49], [50], but the time cost required for computation is relatively small. Compared with the other two networks, the ACB network has lower spatial complexity and requires fewer computer resources.

Compared with ResNe18, the DeepCoral, MRAN, and DSAN have only improved the classification performance on the Hyrank dataset. This shows that these three classical DA methods are not necessarily applicable to the CS-HSIC field, and in practice, whether to apply them should be analyzed according to specific tasks. As a classical DA method for CS-HSIC, TSTnet has shown excellent classification performance on Houston and Hyrank datasets, but its performance on the Pavia dataset is slightly insufficient. In addition, from the classification maps, these DA methods have resulted in greater errors in classes that were originally prone to confusion. The comprehensive classification performance of our network is not inferior to these classical DA methods, but it is far

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TABLE XV
CLASSIFICATION PERFORMANCE OF FOUR NEW NETWORKS ON THE HYRANK DATASET

| Method | DeepCoral(ACB) | MRAN(ACB) | DSAN(ACB) | TSTnet(ACB) |
|---|---|---|---|---|
| OA(%) | 65.36 | 64.36 | **66.75** | 65.70 |
| AA(%) | 44.63 | 41.99 | **50.81** | 42.40 |
| Kappa(%) | 57.28 | 55.90 | **59.53** | 57.69 |
| Parameters(M) | 0.02 | 0.24 | 0.02 | 2.19 |
| Time(s) | 1.8 | 1.9 | 2.1 | 5.0 |

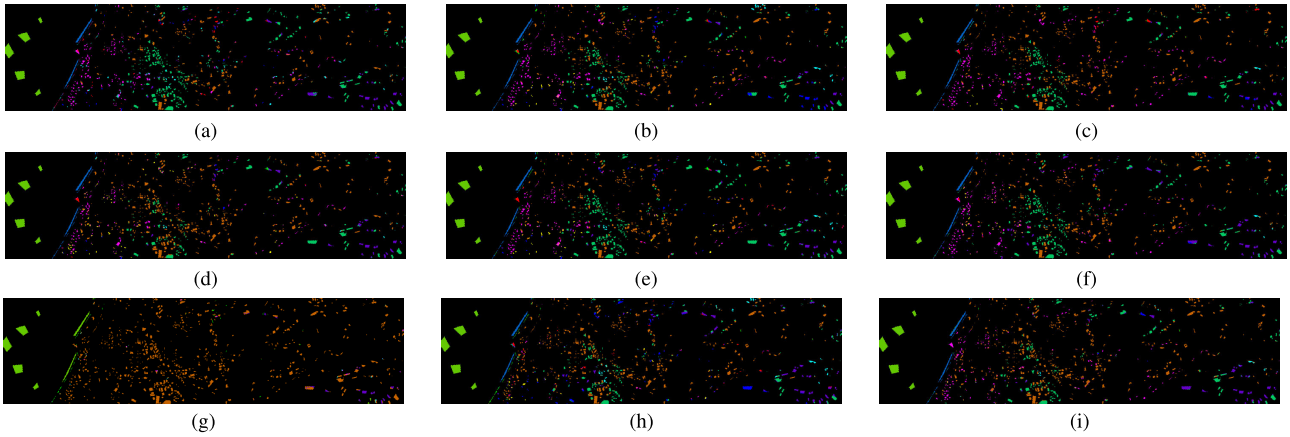


Fig. 15. Classification maps of different methods on the Hyrank dataset. (a) VGG11. (b) ResNet18. (c) DeepCoral. (d) MRAN. (e) DSAN. (f) TSTnet. (g) SDEnet. (h) LDGnet. (i) ACB.

superior to these methods at time complexity and space complexity.

SDEnet and LDGnet are more advanced DG methods in the field of CS-HSIC. Although they have unique advantages, their limitations cannot be ignored. The discriminator of SDEnet tends to binary classification, and LDGnet relies heavily on the quality of text labels. From Tables X–XII and Figs. 13–15 in the case of fewer training samples, the overall classification performance of our network is better than that of these complex networks in different tasks.

Through the above analysis, it is found that the DA and DG methods here are extremely dependent on their own strategies. Our network has better stability and relatively low dependence on training samples. In addition, the time and space complexity is minimal, and it requires far fewer resources than other methods.

### *F. Ablation Study in Combination With Other Networks*

In this section, the four classical DA methods DeepCoral, MRAN, DSAN, and TSTnet combined with ACB network are denoted as DeepCoral (ACB), MRAN (ACB), DSAN (ACB), and TSTnet (ACB). Tables XIII–XV describe the classification performance of new networks.

Through the comparison between Tables X–XII and XIII–XV, it can be found that the overall performance of the new networks is better than the previous networks. On the Pavia dataset, the classification performance has been significantly improved, especially the DSAN (ACB) network with extremely excellent performance. Due to the feature extraction module not extracting features suitable for adaptive computation, many adaptive methods have low accuracy in CS-HSIC. By comparing various new networks with their corresponding original networks on three datasets, most new networks achieve accurate classification at low computational costs.

## V. CONCLUSION

In this article, a lightweight DG network with ACB is proposed. To solve the large parameter number of feature extraction modules and the weak ability to extract domain invariant information, a lightweight cascaded bottleneck structure is proposed to extract domain invariant information. To solve the unreasonable distribution of spectral–spatial features, a multihead attention lightweight auxiliary module is designed, and it rationally allocates weights to optimize spectral–spatial features. On the premise of ensuring good classification performance, the spatial complexity of the feature extraction network is greatly reduced. In addition, this lightweight feature extraction network can be flexibly embedded into other strategies to improve the original network performance. The comprehensive experiments on three cross-scene datasets prove that the whole feature extraction network can effectively extract domain invariant information and optimize spectral–spatial features, and performs well in the DG task. In future work, we will further explore appropriate strategies to combine with the feature extraction network and improve the performance of the CS-HSIC method.

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